E-10-CX-pressure-vorticity v1: does pressure-vorticity coupling give W_- control or new CX? Fast kill audit

ZFL Lab

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MISSION: Find new identity/estimate pressure-vorticity directly controlling W_- or producing necessary consequence C_X of $T^* < \infty$. Single rigorous audit, kill fast if no signal.

INPUT: $\mathcal{C}_{PROVEN} = \{C_E^{weak} : \int_0^t W \to \infty,\ C_{L^3} : \|u\|_3 \to \infty\}$. E-08 obstruction $D_t S \supset -\nabla^2 p$ with $-\Delta p = \partial_i \partial_j (u_i u_j)$. E-09 insufficiency of $\{W, \|\omega\|_2, \|\nabla\omega\|_2, \|u\|_3\}$. Phase E frozen E-05 to E-09 BLOCKED.

FORBIDDEN: $e_i, \lambda_i, \eta_-, D_t e_i$ reuse. Reproducing only Calderon-Zygmund $\|\nabla^2 p\|_{3/2} \lesssim \|\nabla u\|_3^2$ or $\|p\|_{3/2} \lesssim \|u\|_3^2$ without new sign. Only $|W| \le C_S \|u\|_3 \|\nabla\omega\|_2^2$ again. Claiming C_X that is not necessarily divergent under $T^* < \infty$. Hiding nonlocality. $C_E^{weak} \Rightarrow C_E^{strong}$.

AUDIT:

1. Pressure-vorticity identities:

$-\Delta p = \operatorname{tr}[(\nabla u)^2] = \operatorname{tr}(S^2 + \Omega^2) = |S|^2 - \frac{1}{2}|\omega|^2$? Actually $|S|^2 = \operatorname{tr} S^2$, $|\Omega|^2 = \frac{1}{2}|\omega|^2$, $\operatorname{tr}\Omega^2 = -|\Omega|^2$.

So

$$p = (-\Delta)^{-1}\left(|S|^2 - \frac{1}{2}|\omega|^2\right).$$

$W = \int \omega^T S \omega = \int S : (\omega \otimes \omega)$. No p explicit.

But integrate W against p? Consider $W = -\int p$? No.

Alternative: $D_t \omega = S\omega + \nu\Delta\omega$, take divergence of Lamb vector? Use identity $\omega \cdot \nabla p$?

Enstrophy production via pressure: $\int \omega \cdot \nabla \times (u \cdot \nabla u) = W$. Pressure drops from curl.

To get p back, need ∇p coupling: $(u \cdot \nabla)u = \omega \times u + \nabla(\frac{1}{2}|u|^2 + p)$.

Thus $\omega^T S \omega$ is independent of p pointwise, but time evolution of S contains $\nabla^2 p$.

Search for global identity involving $p|\omega|^2$:

$\frac{d}{dt}\int p|\omega|^2$? Then $D_t S$ appears via $\partial_t S$? Compute $\int \nabla^2 p : (\omega \otimes \omega)$?

Note: $W_- = \int (\omega^T S \omega)_- = \int |\omega|^2 (\xi^T S \xi)_-$.

If $S = -\nabla^2 p*$? Not.

Consider representation: $S = \mathcal{R}[u \otimes u]$ via Riesz? Then $W = \int \omega \otimes \omega : \mathcal{R}[u \otimes u]$. This is trilinear form $T(\omega, \omega, u, u)$ with singular kernel. No sign.

Possible C_X: $\int_0^{T_*} \|p\|_{L^{3/2}} dt = \infty$? Since $\|p\|_{3/2} \lesssim \|u\|_3^2$, C_{L^3} gives $\|p\|_{3/2} \to \infty$? If $\|u\|_3 \to \infty$, then $\|p\|_{3/2}$ may diverge, but not necessarily $\int \|p\|_{3/2} = \infty$? Actually if sup diverges, integral may still diverge? Not guaranteed without lower bound.

If $T^* < \infty$, does $\int_0^{T_*} \|p(t)\|_{3/2} dt = \infty$ follow? Need: $\|p\|_{3/2} \gtrsim$? No lower bound from $\|u\|_3$.

Thus pressure alone may not give new divergent quantity.

2. Attempt inequality for W_- via pressure:

Write $W = \int -\operatorname{tr}(S^3) - \frac{1}{4}$... using $\det S$? Known identity $\int |S|^2 = \frac{1}{2}\int |\omega|^2$, but $\int \operatorname{tr}(S^3)$ relates to enstrophy production? Actually $\int \operatorname{tr}(S^3) + \frac{1}{4}\int \omega^T S \omega = 0$ for homogeneous? Betchov relation:

$\langle \mathrm{tr}(S^3)\rangle = -\frac{3}{4}\langle \omega^T S\omega\rangle$ in isotropic turbulence, but not pointwise, only mean with periodic BC and integration by parts.

Global identity (Betchov):

$$\boxed{\int \mathrm{tr}(S^3)dx = -\frac{3}{4}W \quad \text{periodic/homogeneous, EXACT from } \int \partial_j u_i \partial_i u_k \partial_k u_j}$$

Check: $\int \mathrm{tr}[(\nabla u)^3] = 0$, expand $S+\Omega$, gives $\int \mathrm{tr}\, S^3 + 3\int \mathrm{tr}\, S\Omega^2 = 0$, and $\mathrm{tr}\, S\Omega^2 = -\frac{1}{4}\omega^T S\omega$. So $W = -\frac{4}{3}\int \mathrm{tr}\, S^3$.

Thus W_- relates to positive part of $\mathrm{tr}\, S^3$.

But $\mathrm{tr}\, S^3 = 3\det S$ since $\mathrm{tr}\, S = 0$, so $\mathrm{tr}\, S^3 = 3\lambda_1\lambda_2\lambda_3$.

Still sign depends on eigenvalues, forbidden frame? We use integral identity without e_i pointwise but eigenvalues reappear via determinant. Allowed? We forbid e_i pointwise, but global identity may still be usable. However it reintroduces spectral sign indirectly.

Even with Betchov, no separation of W_-.

3. Kill criteria check:

- Only Calderon-Zygmund for p? Yes, we only get $\|p\|_{3/2} \lesssim \|u\|_3^2$, no new sign for W_-. - Only $|W|$ bounds? Yes, trilinear estimate gives again $|W| \le C\|u\|_3\|\nabla\omega\|_2^2$, no improvement. - No C_X necessarily divergent? $\|p\|_{3/2}$ divergence not forced to be integral divergence. - No connection to W_-? Betchov links total W, not W_-.

Thus pressure-vorticity alone does NOT produce inequality of form $W_- \le F$ better than E-09, nor new necessarily divergent C_X stronger than C_{L^3}.

Possible remaining angle: $\int p \det S$ or $\int p|\omega|^2$? Need integration by parts using $-\Delta p = |S|^2 - \frac{1}{2}|\omega|^2$, so $\int p|S|^2$? No sign.

Conclusion for this audit:

$$\boxed{\text{Pressure-vorticity identities: } -\Delta p = |S|^2 - \frac{1}{2}|\omega|^2,\ p = (-\Delta)^{-1}(\dots),\ \int \mathrm{tr}\, S^3 = -\frac{3}{4}W \text{ (Betchov)}}$$

$$\boxed{\text{No new } W_- \le F \text{ beyond } C_S\|u\|_3\|\nabla\omega\|_2^2 \text{ found}}$$

$$\boxed{\text{No } C_X \text{ with } T^* < \infty \Rightarrow C_X \text{ and } C_X \text{ gives contradiction with } C_E^{weak}, C_{L^3} \text{ from pressure alone in this audit}}$$

Classification: BLOCKED/KILLED fast — branch pruned.

$\mathcal{C}_{PROVEN}$ remains $\{C_E^{weak}, C_{L^3}\}$. E-05 to E-09 frozen. E-10 audited and killed.

Next candidate: $\nabla\xi$ with coercive $\nu|\omega|^2|\nabla\xi|^2$ term from vorticity diffusion.